

Q1: [red, blue, blue, red, green, grey]

a)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

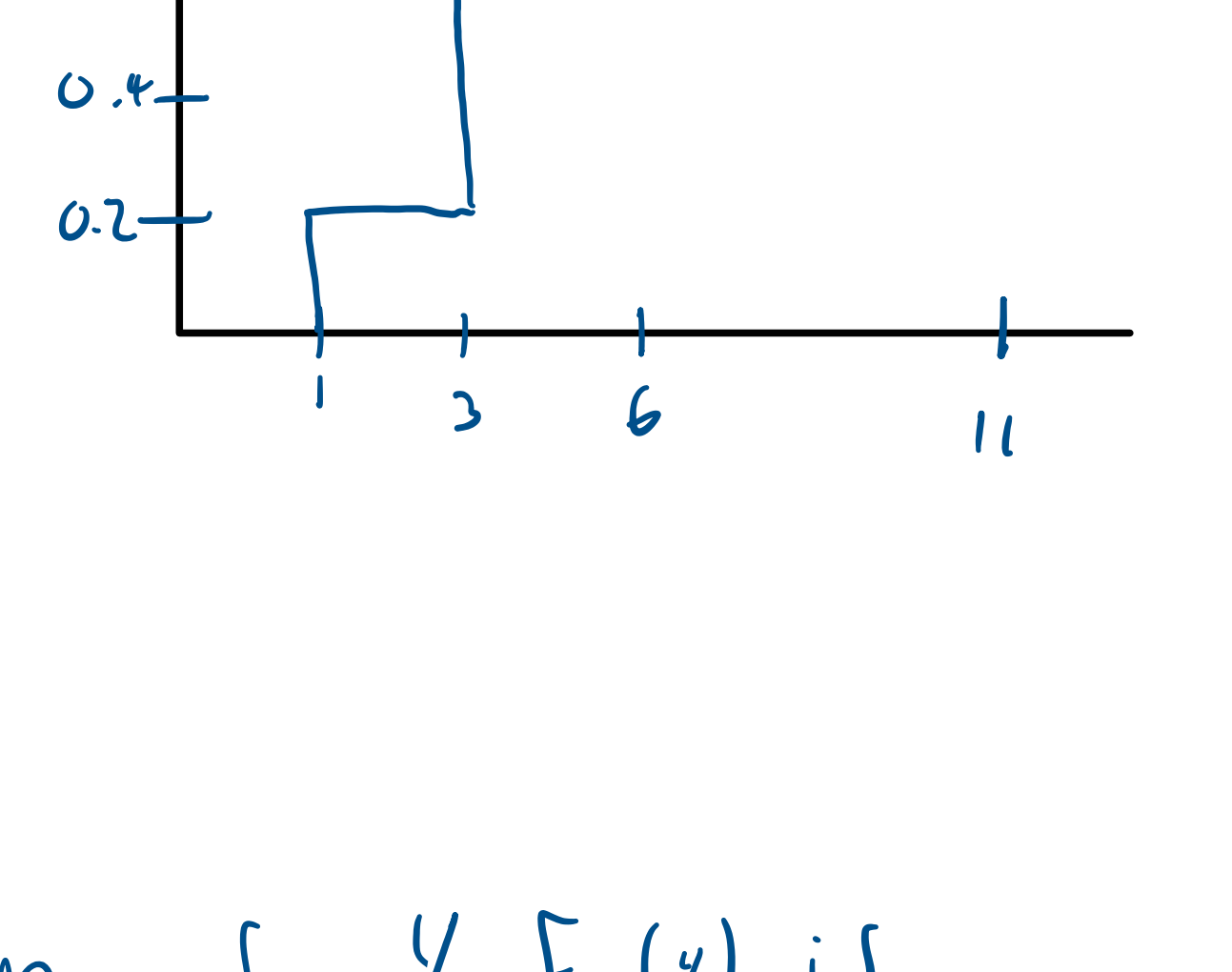
b) Sample proportion:

Red: 0.333
 Blue: 0.333
 Green: 0.167
 Grey: 0.167

c)

$X = [1, 3, 3, 6, 11]$

$$F(x) = \begin{cases} 0 & x \leq 1 \\ 0.2 & 1 < x \leq 3 \\ 0.6 & 3 < x \leq 6 \\ 0.8 & 6 < x \leq 11 \\ 1 & x > 11 \end{cases}$$



Q2.

a)

• The distribution of Y , $F_Y(y)$ is

$F_X(x) = P(X \leq x) \rightarrow F_Y(y) = P(a+bx \leq y)$

$F_Y(y) = P(X \leq \frac{y-a}{b}) = F_X(\frac{y-a}{b})$

• The density is the derivative of the distribution:

$$f_Y = \frac{\partial}{\partial y} (F_X(\frac{y-a}{b}))$$

$$= f_X(\frac{y-a}{b}) \cdot \frac{\partial}{\partial y} [\frac{y}{b} - \frac{a}{b}]$$

Density $= f_X(\frac{y-a}{b}) \cdot \frac{1}{b}$

Sample: $X = [1, 3, 4]$

b) mean $= \frac{1+3+4}{3} = \frac{8}{3} = 2.67$

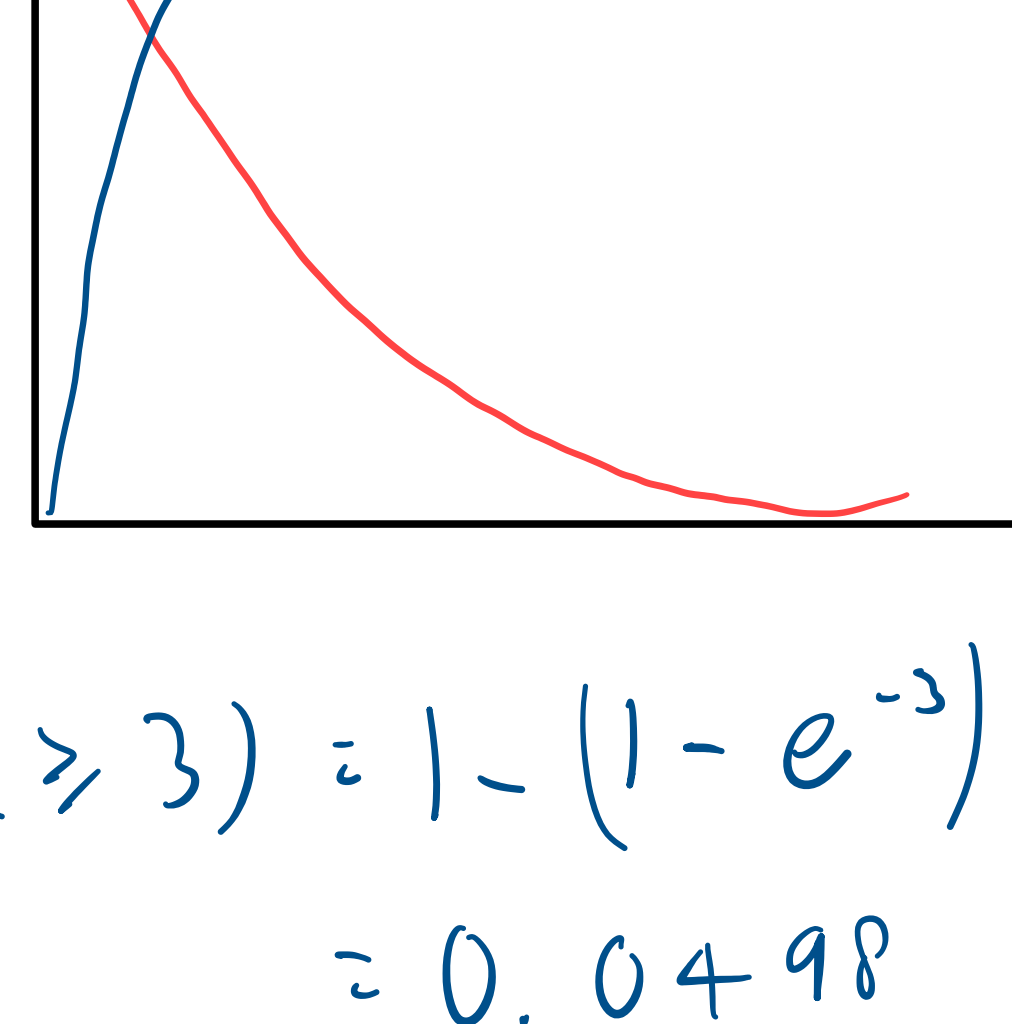
c) $X^2 = [1, 9, 16] = \frac{26}{3} = 8.67$

d) $(2.67)^2 = 7.11$. This is not equal to the mean of part c.

* Nonlinear transformations change moments/statistics in a nonlinear way

Q3. $F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x} & x \geq 0 \end{cases}$

a)



• Distribution
 • Density

b) $P(X \geq 3) = 1 - (1 - e^{-3}) = 1 - 0.95021 = 0.0498$

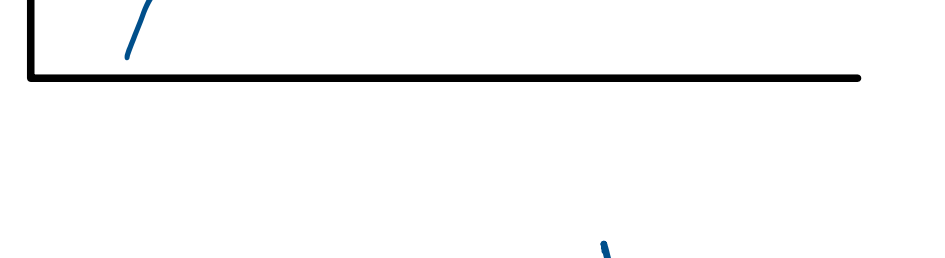
$P(X \leq 35) = 1 - e^{-35} = 1 - 0.000...63 = 0.999...37$

c) $X \rightarrow Y = 1 + 3X$

$X = \frac{y-1}{3}$

$$F_Y(y) = \begin{cases} 0 & \frac{y-1}{3} < 0 \\ 1 - e^{-\frac{(y-1)}{3}} & \frac{y-1}{3} \geq 0 \end{cases}$$

$$F_Y(y) = \begin{cases} 0 & y < 1 \\ 1 - e^{-\frac{(y-1)}{3}} & y \geq 1 \end{cases}$$



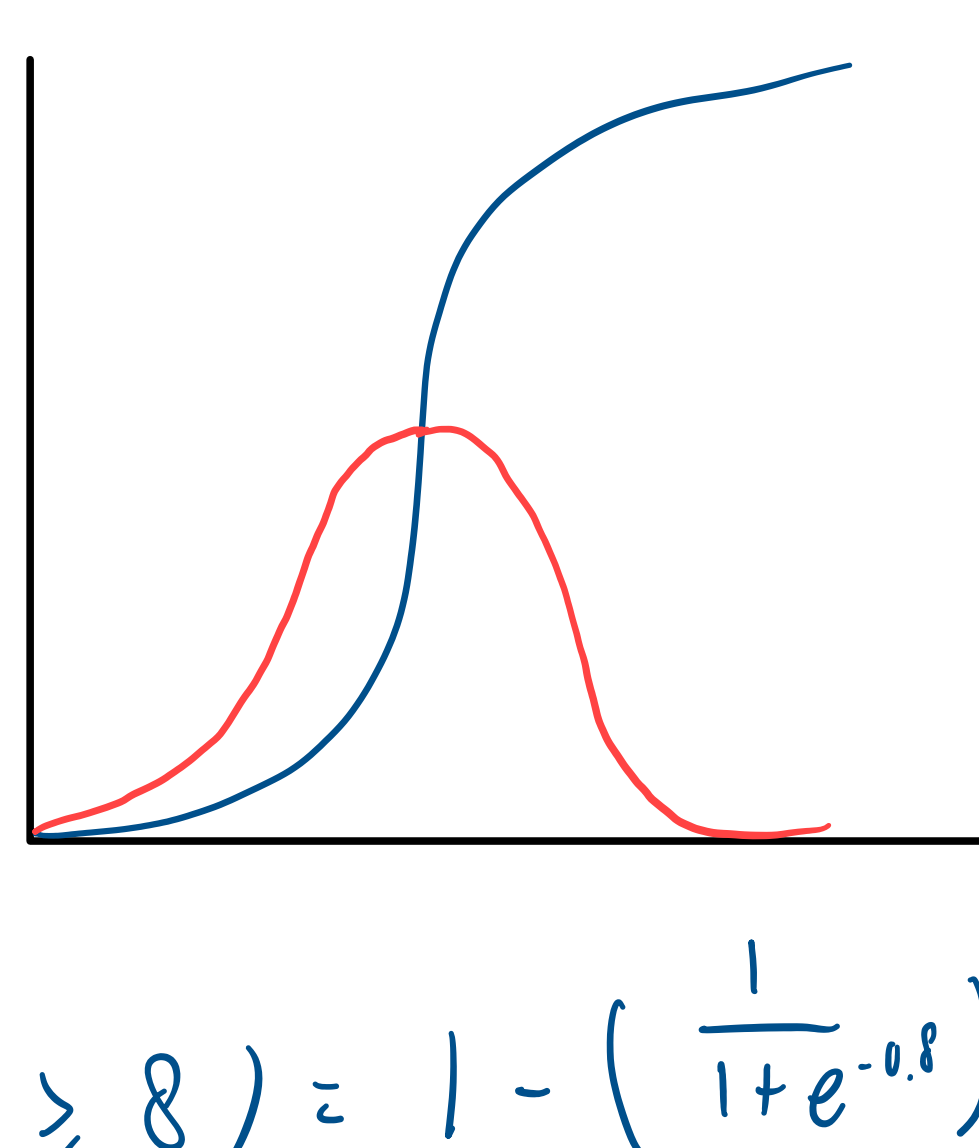
$$P(Y < 10) = 1 - e^{-\frac{(10-1)}{3}}$$

$$= 1 - e^{-3}$$

$$= 0.9502$$

4. $F(x) = \frac{1}{1+e^{-x}}$

a)



• Distribution
 • Density

b) $P(X \geq .8) = 1 - (\frac{1}{1+e^{-.8}}) = 0.31002$

$P(X \leq .3) = (\frac{1}{1+e^{-.3}}) = 0.5744$

c) $X \rightarrow Y = 2X - 1$

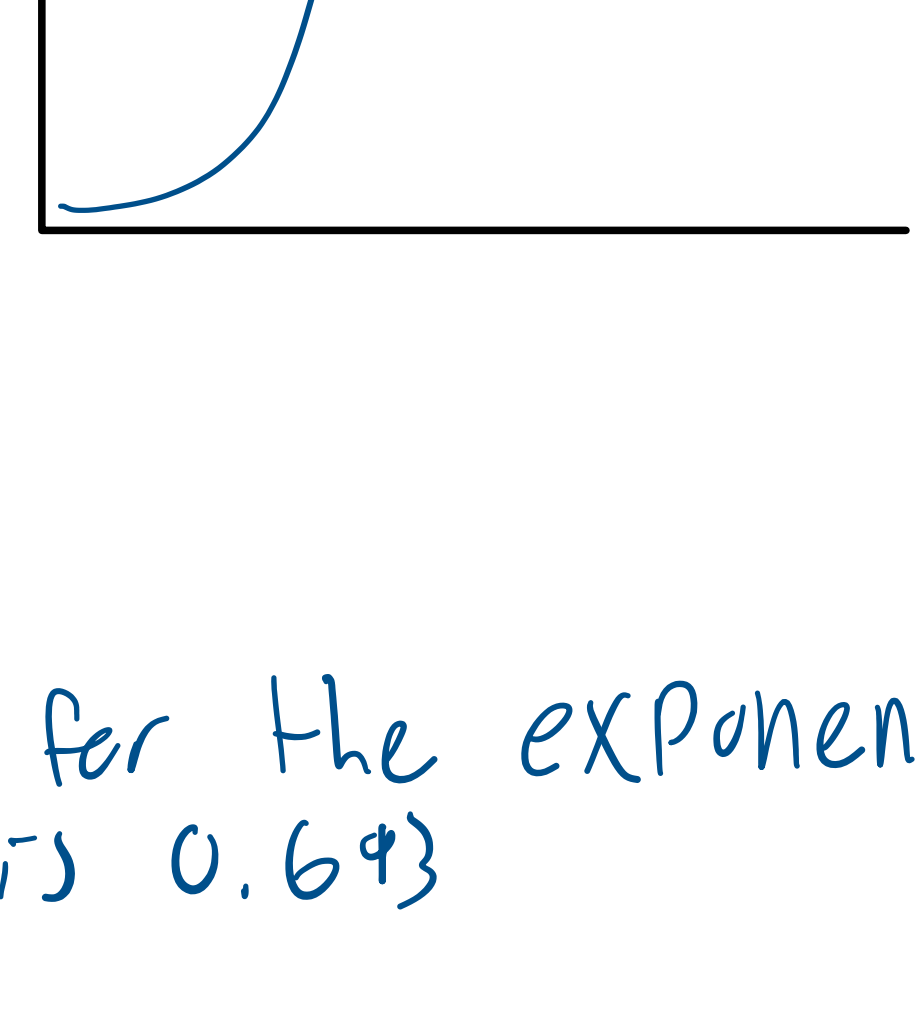
$X = \frac{y+1}{2}$

$F(X) \rightarrow F(Y) = \frac{1}{1+e^{-\frac{(y+1)}{2}}}$

$P(Y < 0) = \frac{1}{1+e^{-.5}} = 0.6255$

graph:

$P(X < 0)$ increases from 0.5 to 0.63.
 Left shift



5.

a) The median for the exponential distribution is 0.693

$0.5 = 1 - e^{-x} \rightarrow (e^{-x} = 0.5) \ln \rightarrow x = -\ln(0.5) = 0.693$

b) The median for the logistic distribution is 0.

c) The quantile function of the exponential distribution is $-\ln(1-u)$

$1 - e^{-x} = u$

$1 - u = e^{-x}$

$\ln(1-u) = -x$

$-\ln(1-u) = x$

d) The quantile function of the logistic distribution is $-\ln(\frac{1}{u}-1)$

$u = \frac{1}{1+e^{-x}} \rightarrow u(1+e^{-x}) = 1$

$e^{-x} = \frac{1}{u} - 1$

$\ln(e^{-x}) = \ln(\frac{1}{u}-1)$

$X = -\ln(\frac{1}{u}-1)$

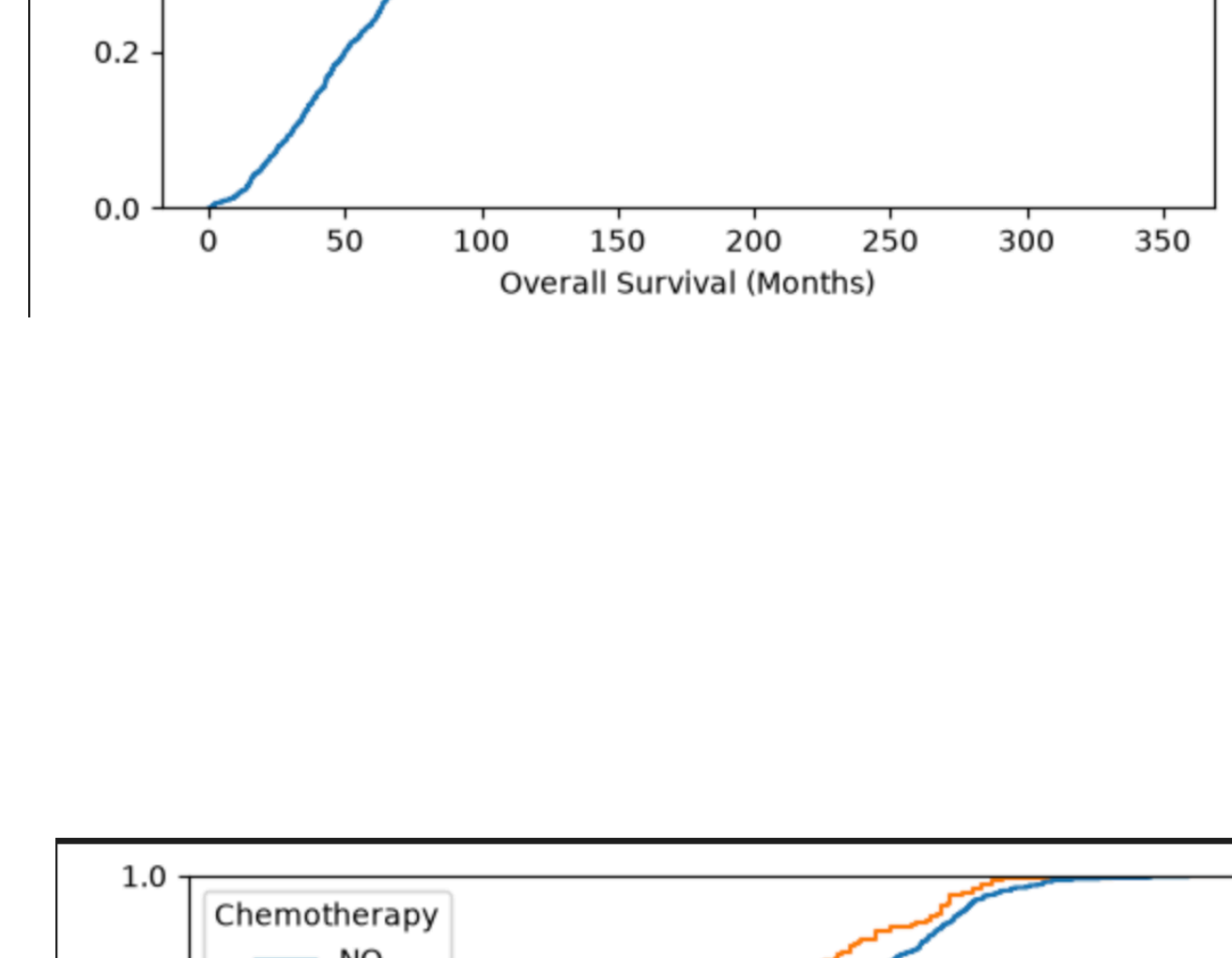
6.

a)

```
import pandas as pd
metabric = pd.read_csv('metabric.csv')

survival = metabric['Overall Survival (Months)']

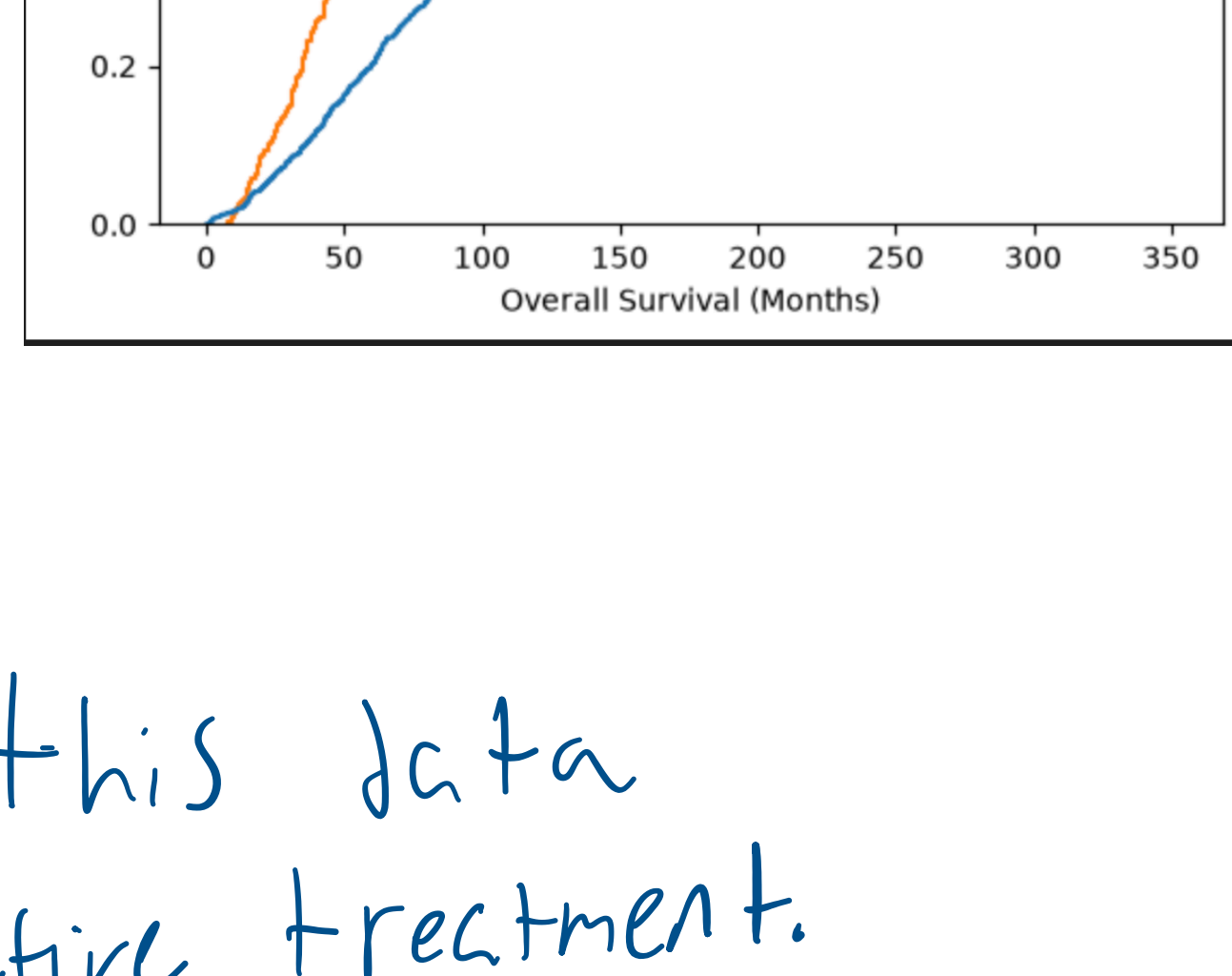
import seaborn as sns
sns.ecdfplot(survival)
```



b)

```
sns.ecdfplot(
    data=metabric,
    x='Overall Survival (Months)',
    hue='Chemotherapy'
)
```

• Conditional on the patient receiving chemotherapy, we would predict that they live a shorter time.



c) It is not clear based on this data whether chemo is an effective treatment.

Patients who undergo chemotherapy tend to have serious forms of cancer, which may be why the life expectancies are shorter with this data. It is important to be carefully when assigning causality to variables.